**JECRC UNIVERSITY, JAIPUR**
**Session: Jan-June 2026**
**Subject: Mathematics-II (DMA025A)**
**B. Tech. II Semester I Year 2025-26**
**Tutorial Sheet–3 (M.M: 64)**

## UNIT II: First order ordinary differential equations

**CO2:** Understand the linear ordinary differential equations along with their applications.

### PART– A (5*2=10)

**A1:** Define order and degree of a differential equation.

**A2:** Write the standard form of a linear first order differential.

**A3:** What is an **integrating factor**? Write the integrating factor for $\frac{dy}{dx} + P(x)y = Q(x)$.

**Ans.** $IF = e^{\int P(x)dx}$

**A4:** Define an exact differential equation and state the condition for exactness.

**A5:** Write the standard form of Bernoulli's differential equation.

### PART – B (3*7=21)

**B1:** Solve the differential equation: $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$. **Ans.** $\boldsymbol{x^2y + xy^2 = c}$

**B2:** Solve the linear differential equation: $\frac{dy}{dx} + y\tan x = \sin x$. **Ans.** $\boldsymbol{y = \cos x\,(\ln|\sec x|) + C}$

**B3:** Solve the Bernoulli's differential equation: $\frac{dy}{dx} + y = y^2$. **Ans.** $\boldsymbol{y = \frac{1}{1+C\,e^x}}$.

### PART – C (3*11=33)

**C1:** Solve the differential equation: $(x^2y + y^3)dx + (x^3 + 3xy^2)dy = 0$. **Ans.** $\boldsymbol{\frac{x^3y}{3} + xy^3 = C}$.

**C2:** Solve the Bernoulli's differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2y^2$. **Ans.** $\boldsymbol{y = \frac{1}{Cx - \frac{x^3}{2}}}$.

**C3:** Solve the linear differential equation: $x\frac{dy}{dx} + y = x\cos x$. **Ans.** $\boldsymbol{y = \sin x + \frac{\cos x}{x} + \frac{C}{x}}$

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